

Module 3: Organic Molecules - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. Why is carbon the backbone of most biological molecules?
2. A. Carbon is the heaviest atom in cells.
3. B. Carbon cannot bond to other atoms.
4. C. Carbon forms up to four stable bonds, building diverse chains and rings.
5. D. Carbon is found only in sugars.
6. Answer: C

7. Why: Carbon's ability to form four stable bonds allows large, varied molecules, which is the basis of life's chemical diversity.

8. Cells join two monomers into a polymer by removing a water molecule. This reaction is called:
9. A. dehydration synthesis
10. B. hydrolysis
11. C. denaturation
12. D. diffusion
13. Answer: A

14. Why: Dehydration synthesis links monomers by removing water. Hydrolysis is the reverse: adding water to break the bond.

15. A protein is heated until it loses its three-dimensional shape and stops working. This change is called:
16. A. replication
17. B. polymerization
18. C. hydrolysis
19. D. denaturation
20. Answer: D

21. Why: Denaturation is loss of shape. Because a protein's function depends on its shape, the protein stops working.
22. Which pairing of molecule and job is correct?
23. A. Lipids store genetic information.
24. B. Nucleic acids store genetic information.
25. C. Carbohydrates always speed up reactions.
26. D. Proteins are the main long-term energy store.
27. Answer: B
28. Why: Nucleic acids (DNA and RNA) store information. Carbohydrates and lipids store energy, and proteins do most cellular work.